

Exponential Equations and an Introduction to Logarithms

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Exponential Equations

① $3^x = 9$

$3^x = 3^2$

$x = 2$

② $8 = 2^x$

$2^3 = 2^x$

$3 = x$

Exponent Property
If $a^x = a^y$
then $x = y$

Solve problem #'s 3-4 using the exponent property shown at the upper right.

③ $6^x = 36$

$6^x = 6^2$

$x = 2$

④ $125 = 5^x$

$5^3 = 5^x$

$3 = x$

Solve the following equations. Write the Roman numeral of each exponent rule that you use.

Equations Requiring One Exponent Rule

⑤ $\sqrt{5} = 5^x$

$5^{\frac{1}{2}} = 5^x$

$\frac{1}{2} = x$

IX

⑥ $2^x = \frac{1}{16}$

$2^x = \frac{1}{2^4}$

$2^x = 2^{-4}$

$x = -4$

VIII

⑦ $3 = 3^{-2x+4}$

$3^1 = 3^{-2x+4}$

$1 = -2x + 4$

$\frac{3}{2} = x$

VI

⑧ $9^x = \sqrt[3]{9}$

$9^x = 9^{\frac{1}{3}}$

$x = \frac{1}{3}$

IX

⑨ $7^x = \frac{1}{49}$

$7^x = \frac{1}{7^2}$

$7^x = 7^{-2}$

$x = -2$

VIII

⑩ $85^{x-2} = 85$

$85^{x-2} = 85^1$

$x - 2 = 1$

$x = 3$

VI

⑪ $32^x = 2^{x+4}$ $(2^5)^x = 2^{x+4}$ $2^{5x} = 2^{x+4}$ $5x = x+4$ $4x = 4$ $x = 1$	⑫ $4^{2x-1} = 64^{5+3x}$ $4^{2x-1} = (4^3)^{5+3x}$ $4^{2x-1} = 4^{3(5+3x)}$ $4^{2x-1} = 4^{15+9x}$ $2x-1 = 15+9x$ $-16 = 7x$ $-\frac{16}{7} = x$	⑬ $1 = 2^{\frac{x}{7}-5}$ $2^0 = 2^{\frac{x}{7}-5}$ $0 = \frac{x}{7} - 5$ $35 = x$
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⑭ $81^x = 9^{x+6}$ $(9^2)^x = 9^{x+6}$ $9^{2x} = 9^{x+6}$ $2x = x+6$ $x = 6$	⑮ $2^{x-5} = 16^{x+1}$ $2^{x-5} = (2^4)^{x+1}$ $2^{x-5} = 2^{4(x+1)}$ $2^{x-5} = 2^{4x+4}$ $x-5 = 4x+4$ $-9 = 3x$ $-3 = x$	⑯ $7^{6x+12} = 1$ $7^{6x+12} = 7^0$ $6x+12 = 0$ $x = -2$
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⑰ $2^{x+5} = 4 \cdot 2^{3x}$ $2^{x+5} = 2^2 \cdot 2^{3x}$ $2^{x+5} = 2^{2+3x}$ $x+5 = 2+3x$ $3 = 2x$ $\frac{3}{2} = x$	⑱ $\frac{7^{2x-3}}{7^x} = 7^{10x+4}$ $7^{2x-3-x} = 7^{10x+4}$ $7^{x-3} = 7^{10x+4}$ $x-3 = 10x+4$ $-7 = 9x$ $-\frac{7}{9} = x$
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⑲ $27 \cdot 3^{4x} = 3^{7+2x}$ $3^3 \cdot 3^{4x} = 3^{7+2x}$ $3^{3+4x} = 3^{7+2x}$ $3+4x = 7+2x$ $2x = 4$ $x = 2$	⑳ $10^{-x+3} = \frac{10^{x-4}}{10^{x+6}}$ $10^{-x+3} = 10^{x-4-(x+6)}$ $10^{-x+3} = 10^{x-4-x-6}$ $10^{-x+3} = 10^{-10}$ $-x+3 = -10$ $-x = -13$ $x = 13$
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III 21) $196 = 7^x \cdot 2^x$ $196 = (7 \cdot 2)^x$ $196 = 14^x$ $14^2 = 14^x$ $\boxed{2 = x}$	IV 22) $16 = \frac{10^x}{5^x}$ $16 = \left(\frac{10}{5}\right)^x$ $16 = 2^x$ $2^4 = 2^x$ $\boxed{4 = x}$	V 23) $4^x = (\sqrt[3]{4})^4$ $4^x = 4^{\frac{4}{3}}$ $\boxed{x = \frac{4}{3}}$
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III 24) $2^x \cdot 5^x = 100$ $(2 \cdot 5)^x = 100$ $10^x = 100$ $10^x = 10^2$ $\boxed{x = 2}$	IV 25) $\frac{56^x}{7^x} = 64$ $\left(\frac{56}{7}\right)^x = 64$ $8^x = 64$ $8^x = 8^3$ $\boxed{x = 3}$	V 26) $\sqrt[5]{7^3} = 7^x$ $7^{\frac{3}{5}} = 7^x$ $\boxed{\frac{3}{5} = x}$
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Equations Requiring Two Exponent Rules

VI 27) $7 = 343^{x+1}$ $7^1 = 343^{x+1}$ $7^1 = (7^3)^{x+1}$ $7^1 = 7^{3(x+1)}$ $7^1 = 7^{3x+3}$ $1 = 3x+3$ $\boxed{-\frac{2}{3} = x}$	I 28) $5^{4x+2} \cdot 5^{3x} = 1$ $5^{4x+2+3x} = 1$ $5^{7x+2} = 1$ $5^{7x+2} = 5^0$ $7x+2 = 0$ $\boxed{x = -\frac{2}{7}}$	II 29) $\sqrt[4]{3} = \frac{3^{x-2}}{3^{5x-4}}$ $\sqrt[4]{3} = 3^{x-2-(5x-4)}$ $\sqrt[4]{3} = 3^{-4x+2}$ $3^{\frac{1}{4}} = 3^{-4x+2}$ $\frac{1}{4} = -4x+2$ $1 = -16x+8$ $\boxed{\frac{7}{16} = x}$
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V 30) $256^{-2x-3} = 4$ $(4^4)^{-2x-3} = 4$ $4^{4(-2x-3)} = 4$ $4^{-8x-12} = 4$ $4^{-8x-12} = 4^1$ $-8x-12 = 1$ $\boxed{x = -\frac{13}{8}}$	I 31) $1 = 6^{2-\frac{3x}{5}} \cdot 6^x$ $1 = 6^{2-\frac{3x}{5}+x}$ $1 = 6^{2+\frac{2x}{5}}$ $6^0 = 6^{2+\frac{2x}{5}}$ $0 = 2+\frac{2x}{5}$ $\boxed{-5 = x}$	II 32) $\frac{2^x}{2^{-x+1}} = \sqrt{2}$ $2^{x-(-x+1)} = \sqrt{2}$ $2^{2x-1} = \sqrt{2}$ $2^{2x-1} = 2^{\frac{1}{2}}$ $2x-1 = \frac{1}{2}$ $\boxed{x = \frac{3}{4}}$
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$$\textcircled{33} 4^x \cdot 5^x = \frac{1}{400}$$

$$\text{III } (4 \cdot 5)^x = \frac{1}{400}$$

$$20^x = \frac{1}{400}$$

$$20^x = \frac{1}{20^2}$$

$$\text{VIII } 20^x = 20^{-2}$$

$$\boxed{x = -2}$$

$$\textcircled{35} \frac{1}{36} = 2^x \cdot 3^x$$

$$\text{III } \frac{1}{36} = (2 \cdot 3)^x$$

$$\frac{1}{36} = 6^x$$

$$\frac{1}{6^2} = 6^x$$

$$\text{VIII } 6^{-2} = 6^x$$

$$\boxed{-2 = x}$$

$$\textcircled{34} \frac{24^x}{4^x} = \sqrt[6]{6^7}$$

$$\text{IV } \left(\frac{24}{4}\right)^x = \sqrt[6]{6^7}$$

$$6^x = \sqrt[6]{6^7}$$

$$6^x = 6^{7/6}$$

$$\boxed{x = \frac{7}{6}}$$

IV

X'

$$\textcircled{36} (\sqrt[4]{3})^5 = \frac{30^x}{10^x}$$

$$\text{IV } (\sqrt[4]{3})^5 = \left(\frac{30}{10}\right)^x$$

$$(\sqrt[4]{3})^5 = 3^x$$

$$3^{5/4} = 3^x$$

$$\boxed{\frac{5}{4} = x}$$

IV

X

Equations Requiring Three Exponent Rules

$$\textcircled{37} \frac{1}{4} = 64^x$$

$$\text{VI } \frac{1}{4^1} = 64^x$$

$$\text{VIII } 4^{-1} = 64^x$$

$$4^{-1} = (4^3)^x$$

$$\text{V } 4^{-1} = 4^{3x}$$

$$-1 = 3x$$

$$\boxed{-\frac{1}{3} = x}$$

$$\textcircled{40} 25^x = \frac{1}{5}$$

$$\text{VI } 25^x = \frac{1}{5^1}$$

$$\text{VIII } 25^x = 5^{-1}$$

$$(5^2)^x = 5^{-1}$$

$$\text{V } 5^{2x} = 5^{-1}$$

$$2x = -1$$

$$\boxed{x = -\frac{1}{2}}$$

VI

VIII

V

$$\textcircled{38} \frac{25^{x+2}}{5^{x+2}} = \frac{\sqrt[6]{5}}{125}$$

$$\text{IV, IX } \left(\frac{25}{5}\right)^{x+2} = \frac{5^{1/6}}{5^3}$$

$$5^{x+2} = 5^{1/6-3}$$

$$5^{x+2} = 5^{-17/6}$$

$$x+2 = -\frac{17}{6}$$

$$\boxed{x = -\frac{29}{6}}$$

IV, IX

II

$$\textcircled{41} \frac{3^6}{\sqrt[4]{3}} = \frac{27^x}{9^x}$$

$$\frac{3^6}{\sqrt[4]{3}} = \left(\frac{27}{9}\right)^x$$

$$\frac{3^6}{3^{1/4}} = 3^x$$

$$3^{6-1/4} = 3^x$$

$$3^{23/4} = 3^x$$

$$\boxed{\frac{23}{4} = x}$$

IV

IX

II

$$\textcircled{39} 1 = (\sqrt[3]{7})^x \cdot 7^{5x-5}$$

$$1 = 7^{x/3} \cdot 7^{5x-5}$$

$$1 = 7^{x/3+5x-5}$$

$$7^0 = 7^{16x/3-5}$$

$$0 = \frac{16x}{3} - 5$$

$$\boxed{\frac{15}{16} = x}$$

X

I

VIII

$$\textcircled{42} \sqrt[5]{2^x \cdot 10^{\frac{x}{5}}} = 1$$

$$2^{x/5} \cdot 10^{x/5} = 1$$

$$(2 \cdot 10)^{x/5} = 1$$

$$20^{x/5} = 1$$

$$20^{x/5} = 20^0$$

$$\frac{x}{5} = 0$$

$$\boxed{x = 0}$$

X'

III

VII

Review on Classifying Numbers

Real Number: A number that is not imaginary (imaginary numbers are numbers that are written with a factor of $i = \sqrt{-1}$, which you will learn later in this course).

Integer: A whole number $(-3, -2, -1, 0, 1, 2, 3, \dots)$.

Natural Number: A whole number greater than zero $(1, 2, 3, 4, \dots)$.

Non-Terminating Decimal: A decimal that continues without stopping ($\frac{1}{3} = 0.3333\dots$ or $0.\overline{3}$).

Non-Repeating Decimal: A non-terminating decimal whose digits display no pattern, i.e. the digits don't repeat ($\sqrt{3} = 1.732050807\dots$).

Rational Number: A number that can be written as a fraction with an integer numerator and an integer denominator ($\frac{2}{3}, -4, 7, \frac{1}{4}$).

Irrational Number: A number that cannot be written as a fraction with an integer numerator and an integer denominator ($\pi, \sqrt{5}, \sqrt{3}/2$).

Irrational numbers are non-terminating, non-repeating decimals ($\sqrt{3}, \pi, \sqrt{7}/5$).

Some rational numbers can be written as non-terminating, repeating decimals ($\frac{5}{6} = 0.83333\dots$ or $0.8\overline{3}$).

An Introduction to Logarithms

Example I (log notation not required): Solve. $2^x = 8$

$$2^x = 2^3$$

The solution is rational. $\Rightarrow$ $x = 3$

Example II (log notation required): Solve. $2^x = 7$

$$x = \log_2 7$$

The solution is irrational. $\Rightarrow$ $x \approx 2.8074$

Notation for Calculating the Value of Exponents

If $b^x = c$, then enter $\log_b c$ into your calculator to find x .

Note: Traditionally, neither "c" nor "b" is allowed to be negative when placed in a logarithmic expression.

Write the solutions of the equations below using " $\log_b c$ " notation. Then use the $\log_y$ button on your iPhone to approximate the answers to the nearest ten thousandth. If you do not have an iPhone, use the calculator on your Windows computer or search "logarithm calculator" on your web browser. Also, there are free graphing calculators on the Google Play Store. Note: Some devices will calculate base ten logarithms ($\log_{10}$) and base "e" logarithms ($\ln$), but no other base logarithms.

④③ $5^x = 9$

$$x = \log_5 9$$

$$x \approx 1.3652$$

④④ $2 = 3^x$

$$\log_3 2 = x$$

$$0.6309 \approx x$$

④⑤ $17^x = 8$

$$x = \log_{17} 8$$

$$x \approx 0.734$$

$$(46) 100 = 91^x$$

$$\log_{91} 100 = x$$

$$1.0209 \approx x$$

$$(49) 8^x = 10$$

$$x = \log_8 10$$

$$x \approx 1.1073$$

$$(47) 7^x = 6$$

$$x = \log_7 6$$

$$x \approx 0.9208$$

$$(50) 5 = 2^x$$

$$\log_2 5 = x$$

$$2.3219 \approx x$$

$$(48) 20 = 6^x$$

$$\log_6 20 = x$$

$$1.672 \approx x$$

Additional Types of Exponential Equations

Determine if you can solve the following equations without a calculator. If you need a calculator, write the solution in $\log_b C$ notation, and then use a calculator to approximate the answer to the nearest ten thousandth. Write the Roman numeral of each exponent rule that you use.

$$(51) 2(6^x) + 9 = 81$$

$$2(6^x) = 72$$

$$6^x = 36$$

$$6^x = 6^2$$

$$x = 2$$

VI

$$(52) -1 = \frac{10^x}{5} - 3$$

$$2 = \frac{10^x}{5}$$

$$10 = 10^x$$

$$10^1 = 10^x$$

$$1 = x$$

$$(53) -8 = -4(8^x) + 4$$

$$-12 = -4(8^x)$$

$$3 = 8^x$$

$$\log_8 3 = x$$

$$0.5283 \approx x$$

$$(54) 1 + \frac{5^x}{2} = 1.5$$

$$\frac{5^x}{2} = 0.5$$

$$5^x = 1$$

$$5^x = 5^0$$

$$x = 0$$

VII

$$(55) 2(9^x) + 6 = 22 \quad (56) 32 = 7(3^x) - 10$$

$$2(9^x) = 16$$

$$9^x = 8$$

$$x = \log_9 8$$

$$x \approx 0.9464$$

$$42 = 7(3^x)$$

$$6 = 3^x$$

$$\log_3 6 = x$$

$$1.6309 \approx x$$